Worksheet — Collisions and Explosions Simulation

Use with *Simulation - Collisions and explosions.html* · conservation of momentum (Block 7): elastic and inelastic collisions, explosions and recoil

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **physics.taneski.com/18-collisions-explosions** in a browser (offline: *Simulation - Collisions and explosions.html*). Choose the type at the top: **Elastic · Inelastic · Explosion** (changing the type keeps the masses and velocities). Pick an **Example** (class demo, slide problems, skaters, cannon) or set the masses and velocities yourself: set a value with ▼ ▲ or type it in the box.

Velocities have a sign: **+ is to the right**. The readouts give *u*, *v* and *p* for A, B and the total, and the kinetic energy before and after. The contact lasts only 0.1–0.2 s, like a real collision; ◄ / ► step through it.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 20 min) · **Part 3** homework (≈ 30 min). The Tasks tab has the same three parts. $\Sigma p_{before} = \Sigma p_{after}$; $E_k = \frac{1}{2}mv^2$ for the questions marked (A.3 link). Answers to 3 s.f. with units.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. 🩹 Cart A (0.25 kg) at 0.50 m s^{-1} sticks to cart B (0.50 kg) at rest. Their common velocity is _____ m s^{-1}.			
2. During that collision, is the force of A on B larger than, smaller than or equal to the force of B on A?			
3. 💥 The same carts start at rest and a spring pushes them apart. Which one moves off faster, and by what factor?			
4. 🧲 Two equal balls, elastic: A at 1.0 m s^{-1} hits B at rest. What happens to A and to B?			

5. (A.3 link) Same carts and speeds: which type loses more kinetic energy — elastic or inelastic (they stick together)?			

6. (slide 14) A 2.0 kg trolley at 3.0 m s^{-1} hits a 1.0 kg trolley at rest and they stick together. Sketch the trolleys before and after (with velocity arrows), then calculate their common velocity. Check: Example → Slide 14: trolleys stick.

Before / after sketch and working

7. (exit question) A 1.2 kg cart at 0.50 m s^{-1} sticks to a 0.80 kg cart at rest. Calculate the common velocity.

Working

Part 2 — During the lab

PASCO Smart Cart demo: three runs. Simulation examples: *Class demo: elastic (magnets)*, *Class demo: Velcro*, *Class demo: explosion* (A = 0.25 kg, B = 0.50 kg, + → right). Press ▶ Start and ● Record for each. Write the class (Capstone) velocities in the grey rows.

run	u_A / m s^{-1}	v_A / m s^{-1}	v_B / m s^{-1}	Σp before / kg m s^{-1}	Σp after / kg m s^{-1}	KE before → after / J (A.3 link)
elastic — simulation						
elastic — class data						
Velcro — simulation						
Velcro — class data						
explosion — simulation						

explosion — class data						

8. For each run, is the total momentum conserved? Is the kinetic energy conserved? (A.3 link) Calculate the percentage of KE lost in the Velcro run.

Working

9. The class elastic run gave v_A = −0.15 m s^{-1} and v_B = +0.32 m s^{-1}, not −0.167 and +0.333 m s^{-1} as in the simulation (an ideal elastic collision). (A.3 link) Calculate the percentage of the kinetic energy the real carts lost. What does this tell you about real magnetic bumpers? Give one other reason why real data differ from the model.

10. In the explosion run, use the *F*–*t* graph (tick Area and Zoom) to compare the impulse on A and on B. Explain the result with Newton's third law.

Part 3 — Homework

11. (slide 14) As in question 6, but now the 1.0 kg trolley moves towards the 2.0 kg trolley at 1.0 m s^{-1}. They stick. Calculate the common velocity.

Working

12. ⛸ A 60 kg skater pushes a 40 kg skater; both start at rest. The 40 kg skater moves off at 1.5 m s^{-1}. Calculate the velocity of the 60 kg skater. (A.3 link) How much kinetic energy did the push give them, and where did it come from?

Working

13. (slide 16) A 500 kg cannon fires a 5.0 kg shell at 120 m s^{-1}. Calculate the recoil velocity of the cannon. Explain why the shell has far more kinetic energy than the cannon although their momenta have the same size.

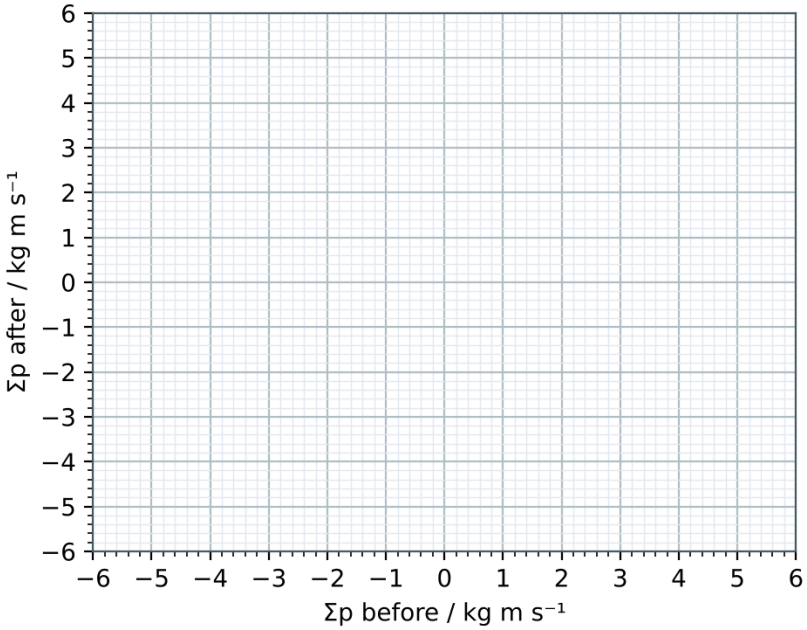
Working

14. A 1.0 kg cart at 2.0 m s^{-1} hits a 3.0 kg cart at rest and bounces back at 1.0 m s^{-1}. Calculate the velocity of the 3.0 kg cart. (A.3 link) Calculate the total kinetic energy before and after the collision. What type of collision is this? Explain.

Working

15. Record at least six trials of different types (some with negative velocities). Copy Σ*p* before and after into the table, plot them and draw the best-fit line.

trial	1	2	3	4	5	6
type						
Σ*p* before / kg m s^{-1}						
Σ*p* after / kg m s^{-1}						



Gradient = _____ . What does it show?

16. Use Newton's third law to explain why the total momentum of two colliding carts does not change.